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Optimizing complex networks for resilience against cascading failure

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Abstract

Our modern society has come to depend on large-scale infrastructure networks to deliver resources to our homes and businesses in an efficient manner. Over the past 10 years there have been numerous examples where a local disturbance has lead to the global failure of systems. In this paper, we use an evolutionary algorithm to evolve complex networks that are resilient to such cascading failure. We then analyze these networks for topological regularities that explain the source of such resilience. The analysis reveals that clustering, modularity and long path lengths all play an important part in the design of robust large-scale infrastructure.
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1. Introduction

Many man-made networks such as the WWW, the Internet and power transmission grids, as well as social networks, financial networks and other organizational and economic infrastructures exhibit similar statistical characteristics [1]. Scale-free networks carry with them a well-recognized strength—tolerance of random failures. But they are particularly susceptible to failure of specific nodes that are highly connected (hubs) and if such removals occur, the networks disintegrate rapidly [2]. When the dynamics of the network are also examined, another weakness is revealed—susceptibility to cascading failures [3].

Cascading failures in power grids are well documented. On August 10, 1996 in Oregon a combination of hot weather and abnormally high-electricity demand caused power lines to sag into trees and trigger a cascade failure of power stations, distribution substations, and assorted other infrastructure which affected power supplies to 11 states [4]. On August 14, 2003 a similar train of events starting in Ohio triggered the largest blackout in North American history [5]. Australia and New Zealand have not been left untouched. In Auckland the failure of four major distribution cables began on January 22, 1998, and when the last of these collapsed almost a month later on February 20 the city was left totally without power. Seventeen days later

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this city had still only managed to regain 40% of its capacity [6]. Where a network is carrying a flow of some particular resource (electricity, gas, data packets, information, etc.) nodes individually experience a load, and in normal circumstances this load does not exceed the capacity of that node. Also, in normal circumstances when flows on the network change, or nodes are added or removed, dynamic adjustment of flows on individual nodes occurs automatically, keeping all nodes loaded below capacity.

Cascade failures are initiated when a heavily loaded node is lost for some reason, and the load on that node (i.e. the flow passing through it) must be redistributed to other nodes in the network. This redistribution may cause other nodes to exceed their capacity causing them also to fail. Even if an overloaded node does not actually fail, protection mechanisms designed into the network may cause it to shut down anyway, in an attempt to prevent node failure. Hence, the number of failed or stressed nodes increases, propagating throughout the network. In particularly serious cases the entire network is affected.

In this paper, we use an evolutionary algorithm to evolve complex networks that are robust to cascading failures. We then apply network statistics to identify topological structures that promote resilience to cascading failure. The following section outlines the model of cascading failures used in this paper. This model serves as our fitness function. Section 3 outlines the evolutionary algorithm used to evolve complex networks. Section 4 details the network statistics used. Section 5 outlines the major results. Finally, Section 6 provides a discussion of the results, and further areas of study.

2. A model of cascading failures

Crucitti et al. [7] (and others e.g. Ref. [3]) proposed a simple model for representing cascading failures on complex networks. They showed that the breakdown of a single node that causes load redistribution to other nodes is sufficient to cause global system failure. This basic model is adopted here as a simulation of cascading failure. Essentially each node is characterized by a given capacity to handle traffic or resource flow. Initially, the network is in a stationary state in which the load at each node is smaller than its capacity. The network is then perturbed, by the "breaking down" of a node. The breakdown changes the balance of flows within the network, leading to the redistribution of load to other nodes. If the extra load cannot be handled by those nodes it is then redistributed to other adjacent nodes. Eventually, the string of failures will result in the reduction in the performance of the entire network. This cascading failure is somewhat analogous to the cascading failures observed in electrical power grids and the Internet. In the remainder of this section we will introduce the cascading failure model.

A generic communication/transport network is represented as a weighted undirected graph G, with N nodes and K edges. G describes an $N \times N$ interaction matrix. If there is an edge between nodes i and j then g_{ij} is assigned a weight in the range (0,1). Otherwise $g_{ij} = 0$. The value of g_{ij} can be thought of as a measure of cost or efficiency of moving between node i and j via g_{ij}. Initially at time $t = 0$, all non-zero g_{ij} values are set to 1, meaning that all paths are functioning equally. The model then iteratively applies a rule for the time evolution of G that mimics the flow redistribution following the failure of a node. To define the network efficiency we assume that the flow between two nodes takes the most efficient (shortest path) connecting them. We use ε_{ij} to denote the efficiency of the shortest path between nodes i and j. In short, the efficiency of the path between nodes i and j is defined as the inverse of the shortest distance: $\varepsilon_{ij} = (1/d_{ij})$, where d_{ij} is the distance between nodes i and j [8]. The average efficiency of a network is then calculated as

$$E(G) = \frac{1}{N(N-1)} \sum_{i \neq j \in G} \varepsilon_{ij}, \tag{1}$$

and is used to measure the efficiency or performance of G at a given time step. The load $L_i(t)$on node i at time t is defined as the total number of shortest paths passing through node i at time t. Each node is characterized by a capacity ψ_i. The capacity is the maximum load that a node can handle before it becomes overloaded and its performance is diminished. The capacity is the number of shortest paths passing through node i at time $t = 0$, multiplied by a tolerance parameter α:

$$\psi_i = \alpha \bullet L_i(0), \quad i = 0, 1, 2, \ldots, N, \tag{2}$$

where $\alpha \geqslant 1$. α can be thought of as a measure of the *stress* that the network is under at time $t = 0$. $\alpha = 1$ means that network is operating at maximum capacity. $\alpha \gg 1$, depicts a network carrying a light load.

The breakdown of a network is simulated by reducing the capacity of a given node i by some margin. For this simulation this margin is set to 75%, (i.e. $\psi_i \times 0.75$). The change in performance of a given node affects the performance of other nodes. Consequently, the redistribution of loads overloads other nodes. At each time step t the following iterative rule is applied to simulate the load redistribution:

$$g_{ij}(t+1) = \begin{cases} g_{ij}(t)\frac{\psi_i}{L_i} & \text{if } L_i(t) > \psi_i, \\ g_{ij}(0) & \text{otherwise,} \end{cases} \tag{3}$$

where j are all those nodes in the local neighborhood of node i.

Fig. 1 illustrates the behavior of the cascading failure model. Fig. 1(A) shows the time evolution of efficiency for three values of α (1.05, 1.1, and 1.35). Fig. 1(B) compares the performance of an Erdös–Rényi (ER) random graph [9], and a network from this study, over tolerance parameters (α values) between 1 and 2. Note that when $\alpha > 1.35$, there is no difference in the performance of these networks.

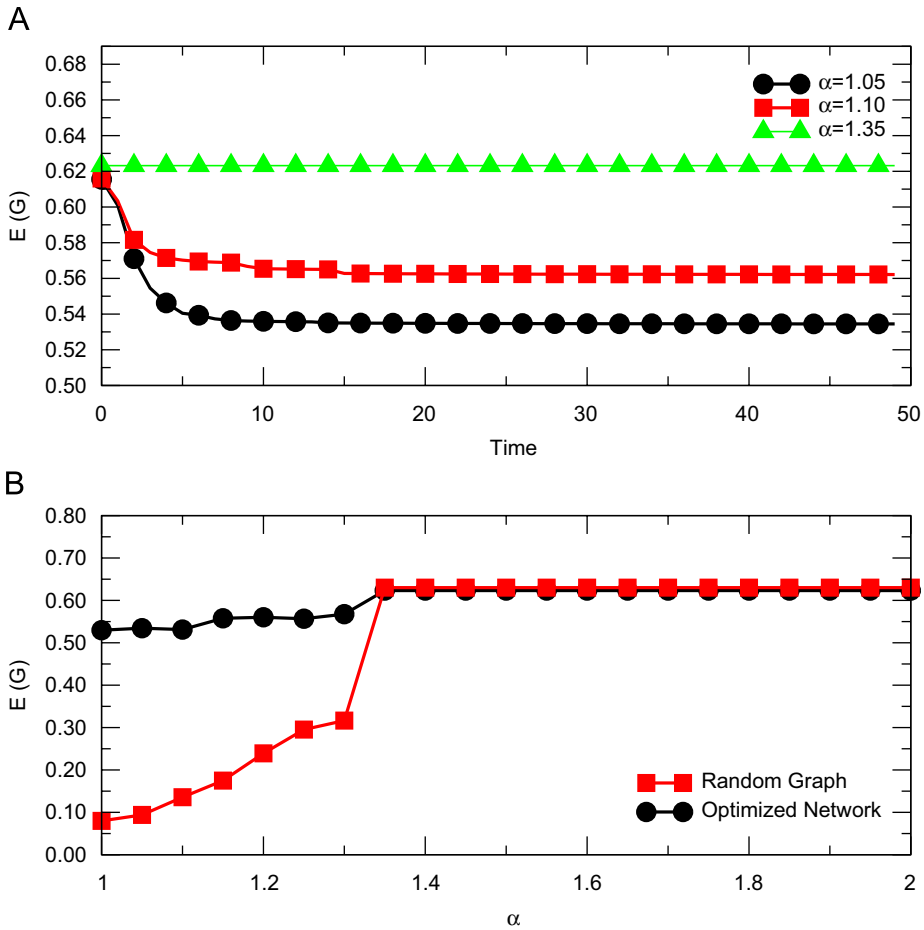

Fig. 1. Behavior of the cascading failure model. (A) Simulation of a cascading failure on a random network with α values of 1.05, 1.10 and 1.35. (B) Comparison of optimized (see Section 3) and random networks containing the same number of nodes and links, for α values of 1–2. Over lower α values of between 1 and 1.35 (highly stressed networks), the optimized network considerably outperforms the random ensembles. However, above this threshold both networks display similar behavior.

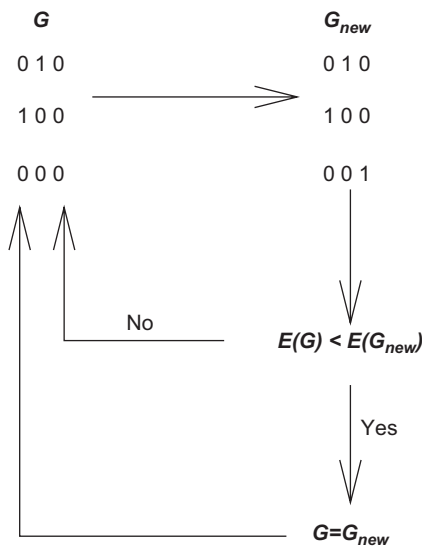

Fig. 2. Basic scheme of the evolutionary algorithm used in this study. Starting from a network/matrix G (here a 3×3 matrix), the algorithm mutates a given matrix entry. The objective function $E(G)$ is then evaluated (see Section 2), and the new network is accepted provided that a greater efficiency is achieved. Otherwise, we start again with the original network G. At the beginning, G is set up with a fixed density of ones.

3. Evolving failure resilient networks

To search for failure resilient networks we have combined the recursive load redistribution algorithm and efficiency measures described in Section 2, with a variation of the Metropolis algorithm [10] or a $1+1$ evolutionary algorithm. The evolutionary algorithm is used to maximize $E(G)$. The α value was fixed at 1.05, to simulate a highly stressed network. The metropolis type algorithm was selected over a population-based approach, because of the computational complexity of evaluating $E(G)$. Fig. 2 shows the basic configuration of evolutionary algorithm used here.

The networks used in this study had a total of 50 nodes. (While being quite small due to computational constraints, this does provide a sufficiently sized network to gain statistically significant attributes). Initially, the network was seeded with 100 edges, connecting the nodes at random. The network was checked to ensure that there was a single-connected component containing all nodes. At each time step the child network was mutated in one of three ways: (1) a new link was added between two previously un-connected nodes; (2) a link connecting two nodes was deleted; or (3) a link connecting nodes i and j was reassigned to connect i and k. These mutation operations were selected without bias. Mutations operations (1) and (2) allow the algorithm to explore configurations with varying degrees of connectivity. They also allow the search to access parts of the search space that may be inaccessible if only operator (3) was available. The new network G_{new}, was evaluated only if: (1) G_{new} contained only one component (i.e. there were no disconnected nodes); and (2) the connectivity of the network was below a critical level c_{critical}. For all experiments, c_{critical} was set at 150 edges, to give a level of connectivity similar to that seen in many natural and physical systems [1]. Also, constraining c_{critical} prevented the algorithm from converging to a trivial completely connected network which was unrealistic. The algorithm was executed for 500 time steps. Experiments with longer runs did not produce quantitatively different results. The topological properties of the final network, G_{final}, were then analyzed. The following section outlines the network properties that were analyzed.

4. Complex networks

Recent studies have shown that many complex networks share common global statistical and structural features (see Ref. [1] for an overview). These properties include scale-free degree distribution, short path length

and high clustering (the so-called "small-world" properties), modularity, and assortativeness. The remainder of this section briefly describes each of these properties.

4.1. Scale-free degree distribution

One of the common structural properties found in many man- made and natural complex networks is a degree distribution with a power-law tail $P(k) \approx k^{-\gamma}$, where exponent γ is in the range between 2 and 3 [1]. The degree of a node is the number of links possessed by that node. Networks exhibiting these power-law degree distributions are known as scale-free networks. Several mechanisms have been proposed for the formation of such topological features. Barabási and Albert [11] showed that a preferential attachment mechanism leads to a degree distribution with a power-law tail. Ferrer-Cancho and Solé [12] showed that the minimization of path length and the number of links contained within a network also leads to scale-free structures. In more recent studies, Brede and Finnigan [13] suggested that scale-free networks possess a number of topological properties that promote stability when confronted with perturbations. Combined, these mechanisms suggest that scale-free networks may be an efficient and stable configuration for many complex networks.

4.2. Small-world networks

Small-world properties can be detected through two statistics, the average shortest path length and the clustering coefficient. The average shortest path length (L) is defined as

$$L = \frac{1}{N(N-1)} \sum_{i=1}^{N} \sum_{j=i+1}^{N} L_{\min}(i,j), \tag{4}$$

where $L_{\min}$ is the shortest path distance between nodes i and j.

Clustering is a common feature found in many networks. The extent of clustering within a network can be quantified by the clustering coefficient (C) [14]. Given a node N_i, with k_i neighbors, E_i is defined to be the number of links between the k_i neighbors. The clustering coefficient is the ratio between the number of links that exist between neighbors (E_i) and the potential number of links $k_i(k_i-1)$ between the neighbors. The average clustering coefficient is

$$C = \frac{1}{N} \sum_{i=1}^{N} \frac{2E_i}{k_i(k_i-1)}. \tag{5}$$

A network is said to have small-world properties if, compared to an ER random graph, the following conditions hold: $L \approx L_{\text{rand}}$ and $C \gg C_{\text{rand}}$.

4.3. Common neighbors and modularity

Many complex systems exploit modularity as a way of coping with complexity. Modules in complex networks are sub-groups of nodes that are highly interconnected within, and loosely connected outside the group. Fig. 3 depicts a social system that is composed of five tightly coupled communities, each of which is loosely connected to the other communities [15]. Pimm [16] proposed that the ratio between the number of neighbors nodes i and j have in common divided by their total number of neighbors (S_{ij}) as a simple measure of system modularity or compartmentalization. The average modularization M is the average of S_{ij} across all nodes:

$$M = \frac{1}{N(N-1)} \sum_{i=1}^{N} \sum_{\substack{j=1 \\ j \neq i}}^{N} S_{ij}. \tag{6}$$

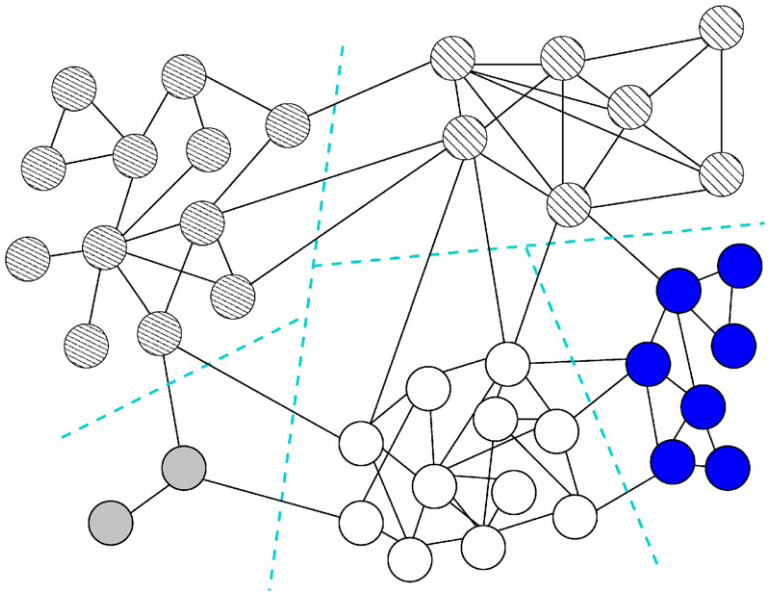

Fig. 3. A Complex network with five distinct modules. This figure shows a social network comprising 38 nodes. The system is self-organized into a number of highly cohesive sub-groups (shading is used to depict members of the same group), that are highly coupled; with loose connections between groups (the loose couplings are those edges that cross the dashed lines).

4.4. Assortativeness

A network displays assortative mixing if the nodes in the networks that have many connections tend to be connected to other nodes with many connections. A network is said to be disassortative if the highly connected nodes tend to be connected to nodes with few connections. The degree of assortativeness can be detected through the use of the Pearson's correlation coefficient. Such correlations can be defined as

$$\Gamma = \frac{c\sum_i j_i k_i - \left[c\sum_i \frac{1}{2}(j_i + k_i)\right]^2}{c\sum_i \frac{1}{2}(j_i^2 + k_i^2) - \left[c\sum_i \frac{1}{2}(j_i + k_i)\right]^2}, \tag{7}$$

where j_i and k_i are the degrees of the vertices at the ends of the ith edge. The constant c is defined as the reciprocal of m where m is the number of edges i.e. $c = 1/\mathrm{m}$. A network displays assortative mixing when $\Gamma > 0$ and disassortative mixing when $\Gamma < 0$. Studies have found that social networks display assortative mixing, while systems with a power-law degree distribution are disassortatively mixed [15].

5. Result

The evolutionary algorithm outlined in Section 3 was executed 100 times. In Section 5.1, we analyze the topological properties of the resultant network. In that section, we also examine how the statistical features of the networks change as they are optimized. Section 5.2 examines the degree distribution of the optimized networks. Finally, section 5.3 compares the statistical characteristics of the optimized networks with those of two random null models.

5.1. Network evolution

Fig. 4 shows the improvement of the network efficiency ($E(G)$) over time, together with three snapshots of the network at time steps 0, 250, and 500. The figure illustrates that over time the network becomes more

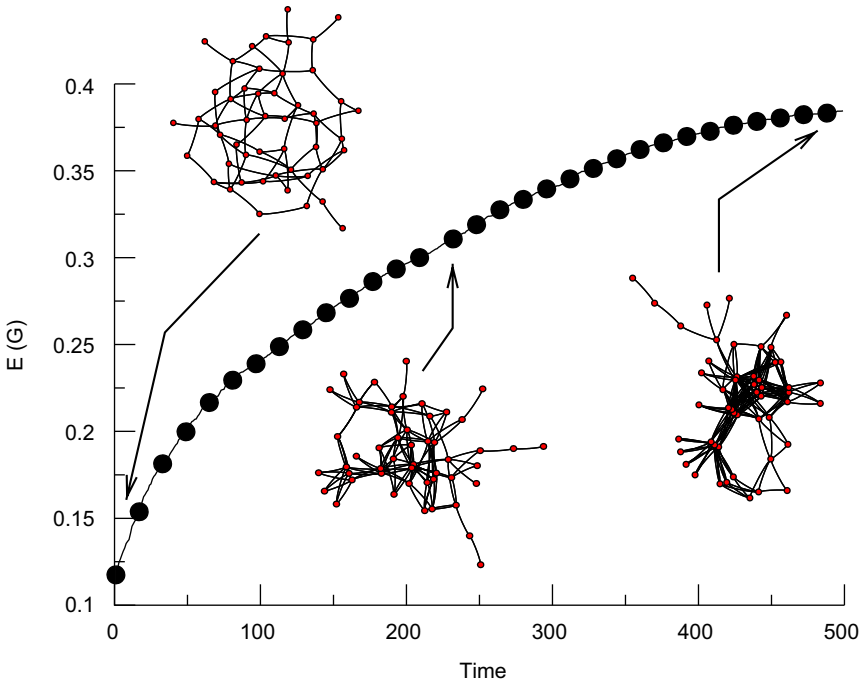

Fig. 4. Progression and emergence of structure. Initially at $t = 0$ the network resembles a random graph. As the network becomes more robust structures such as hubs, star-like sub-structures and highly connected components begin to emerge ($t = 250$). Finally at $t = 500$ the network is highly optimized and displays some decisive structural characteristics, such as modules, and hubs.

stable, and structure—emergence of hubs, clustering, bi-partite sub structures, and modules—begins to occur. Fig. 5 shows the evolution in five different network characteristics for a typical run.

5.2. Degree distribution

Extensive studies of the degree distribution of real-world networks has identified three main classes of networks: (1) scale-free networks, characterized by a vertex connectivity distribution that decays as a power law; (2) broad-scale networks, characterized by a connectivity distribution that has a power-law regime followed by a sharp cut-off; and (3) single-scale networks, characterized by a connectivity distribution with a fast decaying tail [17]. Fig. 6 shows the cumulative degree distribution for the networks evolved in this study. This suggests our evolved networks to the broad-scale networks. Despite the short tail (which could be due to finite size effects or the random initial conditions), there is a significant fraction of nodes with degrees larger than 20 while the average degree is approximately 6. This indicates that the resulting network is quite inhomogeneous.

5.3. Comparison

In previous studies [1,2,11,18], it has been highlighted that many of the statistical properties of a network are derived directly from the degree distribution. In an attempt to determine how "unique" or "special" these evolved networks are we have compared their network statistics to those of two random null models. The first of these is an ER random graph to determine those characteristics which can be accounted for purely by random interactions. The second model is the degree randomization model as described by Ref. [18]. This model assumes that the degree distribution is the source of the network properties. The model randomizes node connection (i.e. which node is connected to which other node), but preserves the individual node degree characteristics. Comparison between the evolved networks and the two null models, shows what is unique to

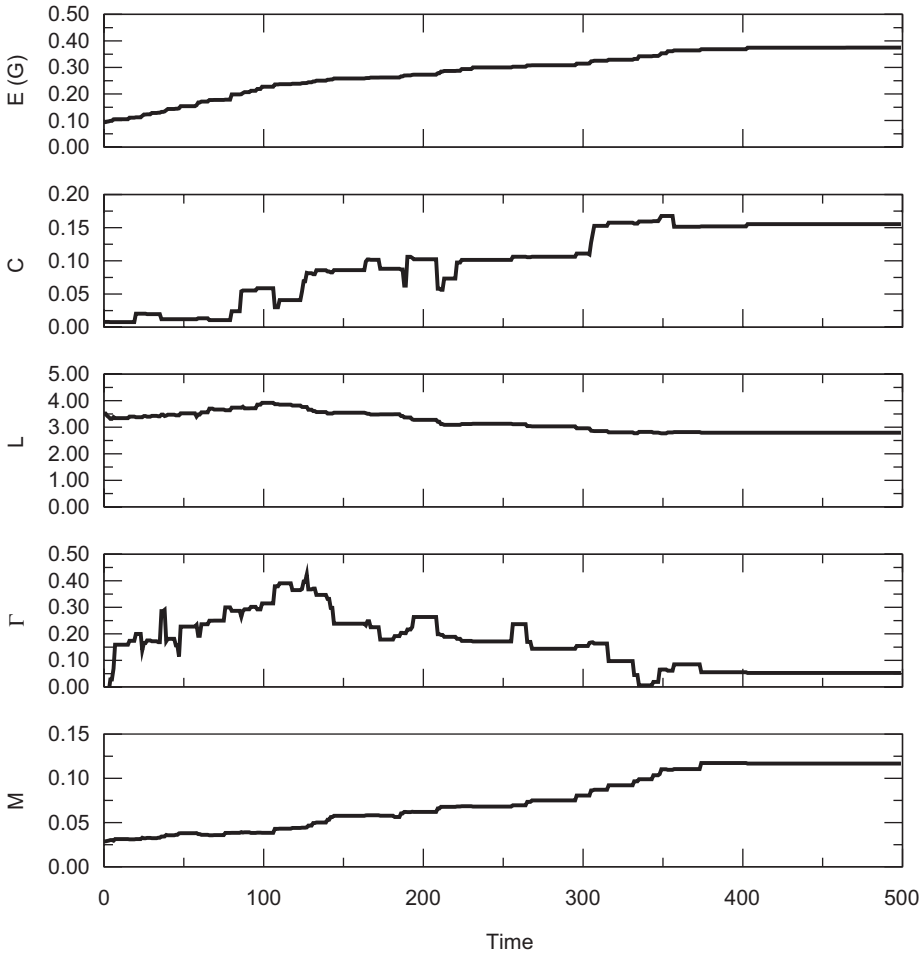

Fig. 5. Changes in network statistics during a typical run. As can be seen the efficiency $E(G)$ of the network when subjected to attack improves over time. The network also becomes more clustered and modular over time (C and M), while the average shortest path length (L) slowly decreases. Assortativeness (Γ) increases initially and then decreases.

the evolved networks, as well as what properties can be accounted for by random assemblage and the degree distribution.

Fig. 7 shows a comparison between the evolved networks and the two null models. Fig. 7(A) shows that the variation in clustering can be best accounted for by the degree distribution. The path length (Fig. 7B) of the evolved networks is somewhat higher than that of the ER random graphs and the randomized degree networks. This characteristic can be considered to be unique to these networks. The wide variation in the assortativeness of the networks (Fig. 7C) indicates that this characteristic is not an important one in the formation of cascade resilient networks. Finally, the degree of modularity (Fig. 7D) in the evolved networks is higher than that found in the random null models. Although the degree distribution does account for some of the observed clustering and modularity, it does not account for all the observed features. The evolved networks have a somewhat "unique" wiring which gives them increased modularity, and clustering than what is given solely by their degree distribution.

6. Discussion

In this paper, we have employed the use of an optimization and search algorithm to identify network characteristics that seem to be associated with robustness against cascading failures. Fig. 4 illustrates that the

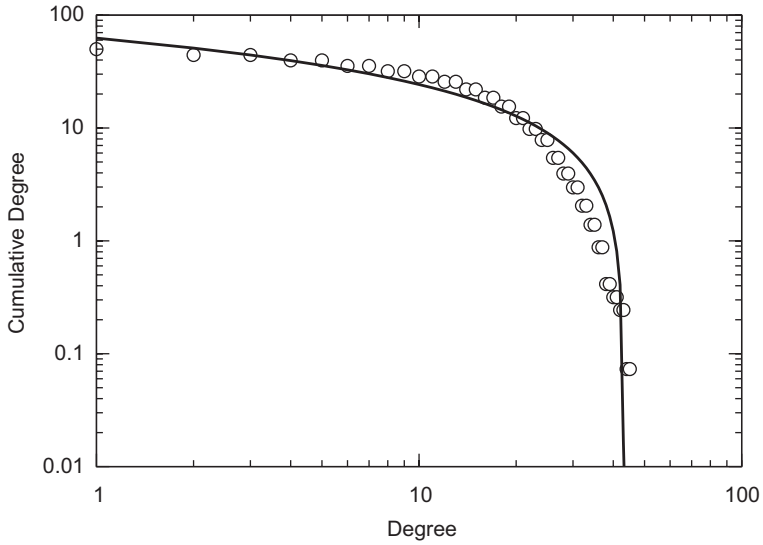

Fig. 6. Cumulative degree distribution for the evolved networks. The fast decaying tail is an indication of a single-scale network.

optimized networks display a degree of structural regularity. One of the most striking characteristics of the resulting networks is their modular design. Self-contained or modular components are often used as a means to isolate failures, and reduce the overall complexity of the system. We suggest that modularity in complex networks could also be serving a similar function. Local failures could be propagated locally and resolved, thus affecting only a small part of the network. Clustering also appears to be an important factor. High clustering provides a series of alternative pathways through which flows can pass, thus avoiding the failed component. The results here show that the degree distribution is an important factor in promoting modularity and clustering. That is, the way links are allocated to nodes drives the modular and clustered nature of the resulting network.

One of the more interesting findings is that the optimized networks have longer path lengths than the two comparative random null models. The existence of long path length means that a disturbance needs to pass through a greater number of intermediate steps (and can potentially be resolved at each step), before the entire network is exposed to the failure. The longer the delay the greater the chance the network will resolve the perturbation. The placement of hubs and highly connected components is crucial in maintaining resilience (as the balance between modularity, clustering and path length needs to be maintained).

Hubs play an important role in many large scale infrastructure networks. While the variation in assortative mixing (see Fig. 7C), and finite size effects make it difficult to determine the role and configuration of hubs that makes networks stable, we postulate that collections of interconnected hubs (hub ensembles) allow perturbations to be distributed and reabsorbed quickly. Visual inspection of the optimized networks shows that this is a common feature; however a systematic test needs to be developed to gain a full understanding of the interconnected nature of the hubs.

Many biological, social and large scale infrastructure networks display a surprising degree of similarity in their overall organization. Although these systems may look structurally similar, the origins of the similarity may be quite different. Biological networks, for example, exploit homeostasis provided by certain network properties, while technological networks arrive at the same properties as the result of a trade-off between communication efficiency and link cost [12,19]. For the simple system dynamics studied here, we suggest that modular design and clustering are key properties when designing complex networks that need to be robust to cascading failure. But, it should be noted that the mere observations of increased clustering and modularity does not imply robustness to cascading failure.

Finally, the work presented here opens a number of additional lines of study, and three deserve mention: (1) The networks studied here are all homogeneous. How does the system organize itself, where certain key components are more likely to have a major failure? (2) If capacity of a given node is measured in terms of a

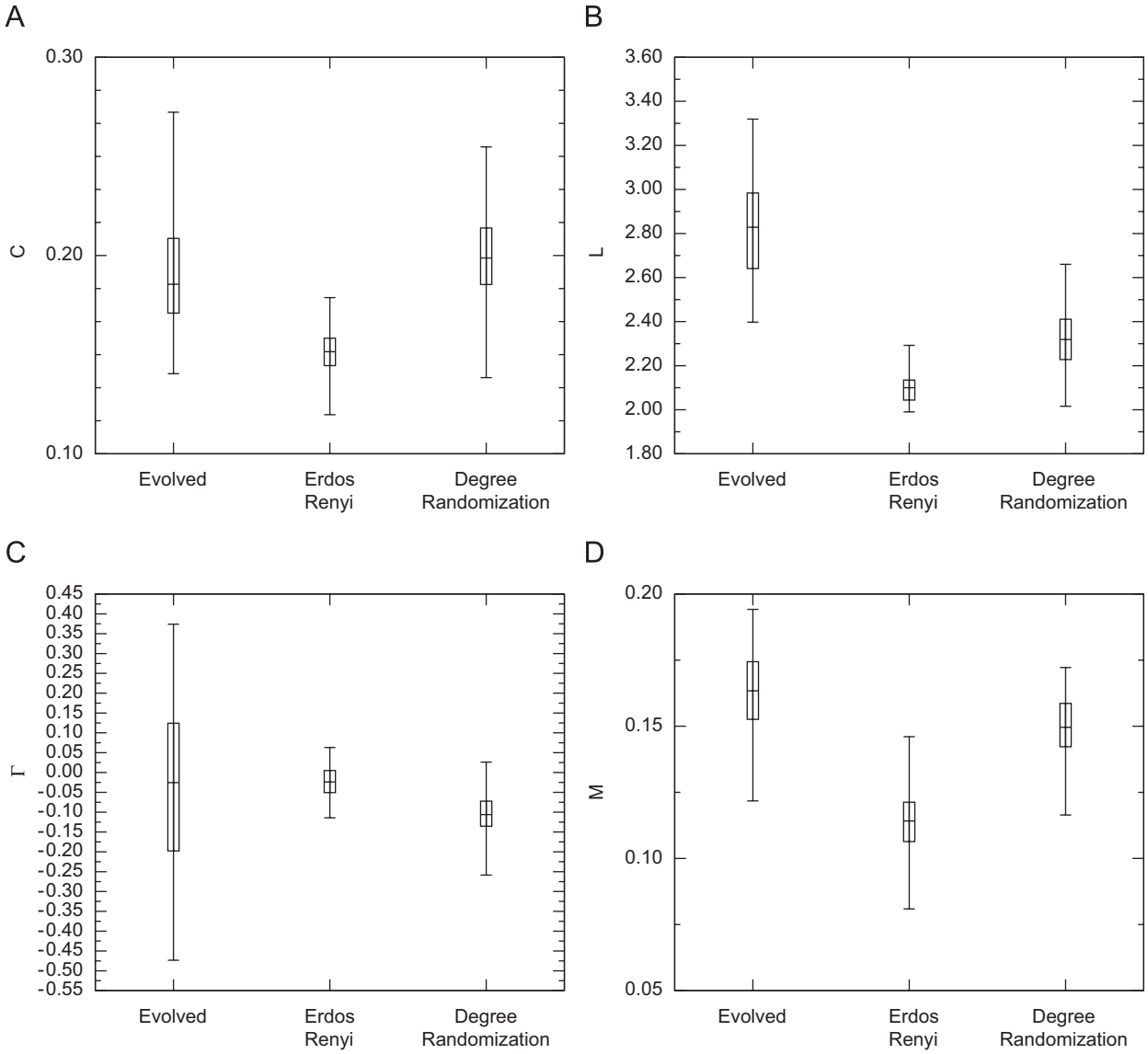

Fig. 7. Comparison between the evolved networks and the two random null models. (A) Clustering coefficient (C); (B) path length (L); (C) assortativeness (Γ); and (D) modularity (M).

cost function, what configuration generates the most robust topology, while minimizing cost? (3) Many natural systems display a high degree of resilience to cascading failures. How do these networks compare with the evolved networks, and to large scale infrastructure networks? What are the sources (reasons) for the variations? All of these questions require further experimentation but can be explored in the context of the framework proposed here.

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